

Adèle DEJOIE

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Research Profile

I am a fourth-year student at ENS Rennes and a graduate of the MVA Master's program, seeking a PhD starting in Fall 2027.

I am broadly interested in challenging research problems in deep learning, with particular interests in foundation models, model architectures, representation learning, and reinforcement learning.

Education

ENS Paris-Saclay – Paris-Saclay University

Sept 2025 – Sept 2026

MSc, MVA: Mathematics, Vision, Learning - Graduated with Highest Honors

- One year of advanced coursework in modern machine learning and its mathematical foundations, taught by leading experts in the field.
- Most courses include an independent research project carried out in pairs or small groups, fostering autonomy, collaboration, and scientific rigor.
- **Coursework:** Deep Learning; DL Theory; Reinforcement Learning; Probabilistic Graphical Models & Deep Generative Models; ML for Text and Graph Data; Speech and NLP; Multimodal XAI; Graphs in ML; Convex Optimization; ML for Time Series.

ENS Rennes

Sept 2023 – Aug 2027

& Magistère in Mathematics – University of Rennes

Sept 2023 – July 2025

- M1 in Fundamental Mathematics; BSc in Mathematics; BSc in Computer Science.

Lycée Henri-IV, Paris

Sept 2020 – July 2023

Scientific Preparatory Classes

- Three years of intensive post-secondary education preparing for entrance exams to France's top engineering and research schools, with advanced coursework in Mathematics, Physics, and Computer Science.

Research Experience

Incoming Research Intern

Tokyo, Japan

RIKEN AIP - Tensor Learning Team - Qibin Zhao

Jan 2027 – July 2027

- Research on tensor-network methods for LLM compression, with the goal of reducing model size while preserving performance.

Research Intern

Bordeaux, France

INRIA - Flowers Team - Pierre-Yves Oudeyer & Loris Gaven

May 2026 – Nov 2026

- Research on LLM-based Knowledge Tracing and student answer prediction.
- Designed compact history representations and model-agnostic conditioning mechanisms enabling LLMs to efficiently exploit structured student interaction histories.
- Optimized the training pipeline for efficient multi-GPU training.
- Outperformed published state-of-the-art models across challenging generalization settings, while retaining efficient inference.
- Current work explores reinforcement learning with reasoning-capable LLMs to unify Knowledge Tracing and answer prediction within a single model. Manuscript in preparation, targeting ICML 2027.

Research Intern

Osaka, Japan

Osaka University - Jun Sekine

June 2025 – July 2025

- Studied optimal stopping problems using stochastic calculus, with a focus on the analysis and optimization of stopping strategies.

Research Intern

Lille, France

INRIA - Magnet - Damien Sileo

June 2024 – July 2024

- Designed, generated, and refined Bayesian networks to create valid probabilistic problems under minimal assumptions. Used the generated dataset to fine-tune a BERT-based language model.

Research Projects

Selection Dynamics in Digital Red Queen

2026

- Challenged a subtle assumption in Digital Red Queen [Kumar et al., 2026] by asking whether multiplayer and pairwise evaluation induce the same selection pressure.
- Turned this question into a complete experimental study in roughly eight days, from analysing DRQ and its codebase to designing controlled experiments, running them, and writing the final manuscript.
- Uncovered and analyzed non-trivial phenomena shaping the evolutionary selection of warriors, motivating further investigation in this direction.

Molecular Graph-to-Text Generation

2026

- Investigated different ways of translating structured molecular graphs into natural language, developing both end-to-end graph-conditioned language models and contrastive retrieval-based approaches.
- Analyzed the strengths and failure modes of these approaches, showing how pretrained language knowledge and graph-derived structural information complement each other in molecular description generation.

Research Outputs

Dejoie, A. (2026). *Shandess: Adapting Large Language Models for Student Response Prediction*. MVA research internship report, Inria Flowers. [Report]

Dejoie, A. (2026). *What Does Digital Red Queen Select For? A Compute-Efficient Study of Pairwise vs. Multiplayer Selection*. Research manuscript. [Manuscript]

Dejoie, A. & Turnel, V. (2026). *Molecular Graph-to-Text Generation*. MVA research project report. [Report] [Code]

Dejoie, A. (2025). *Study of the Optimal Stopping Problem using Stochastic Calculus*. Osaka University research internship report. [Report]

Dejoie, A. (2024). *Generation and Fine-Tuning of Probabilistic Problem Sets using Bayesian Graphs*. Inria Magnet research internship report. [Report]

Technical Skills

Machine Learning: PyTorch, Hugging Face Transformers, PEFT/LoRA, distributed and multi-GPU training.

Programming & Tools: Python, Java, C++, C, SQL, Linux, Git, Slurm, LaTeX.

Languages

French: Native **English:** Fluent (C1) **Japanese:** Beginner